Education to Eternity

JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
An Autonomous Institute, with NAAC "A" Grade
Basic Science & Humanities Department
Semester-II

MSE

Course: B. Tech.
Subject: Advanced Chemistry.
Subject Code: CSE2T002/AI2T002/DS2T002
Year/Semester: Semester II (First Year)
Max Marks: 20 **Date: 15/04/2024** **Duration:- 1 Hr**

Instructions to the Students:
1. **All the questions carry marks as indicate.**
2. **Assume suitable data whenever necessary.**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**

Students should be able to:

CO1- Describe various properties of water, Describe types of fuel, refining of Petroleum, orbital, electronic configuration , nonmaterial and polymers, energy levels, spectroscopic technique

CO2- Interpret the various classification of fuel, various sources of water , refining of petroleum, classification of CNT , various properties of nonmaterial and polymers, ionization energies

CO3- Apply the Knowledge of characteristics of good fuel, Synthesis of nonmaterial, liquid crystal polymers, zeolite process, Ion exchange process, Hot Lime –Soda process, acid base concept, spectroscopic techniques

CO4- Analyze the question on Proximate and Ultimate analysis of coal, potential use of nonmaterial, phases of thermotropic polymers , analyze question on water characteristics

CO5- Estimate a Modal on commercial grading of coal, synthesis of nonmaterial, advanced polymers, spectroscopic technique , doping

CO6- Organize coal, water as per quality, energy level diagram of diatomic molecules, nonmaterial and polymers.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

	QUESTIONS	Level/CO Mapped	Marks
Q.1	Define the principle of the Lime Soda Process.	1/CO1	1
Q.2	Name the salts responsible for Temporary Hardness in Water.	1/CO1	1
Q.3	Define nano-scale.	1/CO2	1
Q.4	What is nanomaterial?	1/CO2	1

SECTION B

1. **All Questions are compulsory.**
2. **Solve answer in brief (two, three Sentence)**

Q.5	Categorize the types of nanotubes.	2/CO2	2
Q.6	Name the types of Hardness with its salts in Water.	2/CO2	2
Q.7	Classify and explain the Zeolite Process.	2/CO3	2
Q.8	Illustrate and explain the application of MWCNTs.	2/CO2	2

SECTION C

1. **Solve any two questions.**

Q.9	Using the information provided, calculate the hardness of the 900 liter water sample that was completely softened by a zeolite softener and required 100 liters of NaCl solution with a concentration of 200 mg/l of NaCl for regeneration.	3/CO3	4
Q.10	Analyze how nanomaterials are used in sensors and discuss their practical applications.	4/CO2	4
Q.11	Evaluate and discuss the uses of nanomaterials in the Catalysis.	4/CO2	4

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